

Why Doesn't This Texture Feel Right?

A Hands-On Workshop on Data-Driven Haptic Texture Reproduction and Typical Pitfalls

Yasemin Vardar, Koen Wosten, Jagan K. Balasubramanian, Umut Kenanoglu, Paulina Gallego, and Virginia Civerati

Where: Room C, Santa Maria della Scala, Siena, Italia

What you will **learn** in this workshop:

- Record and render tactile textures
- Mitigate common pitfalls affecting quality of recorded and rendered signals
- Use SENS3 dataset for your research

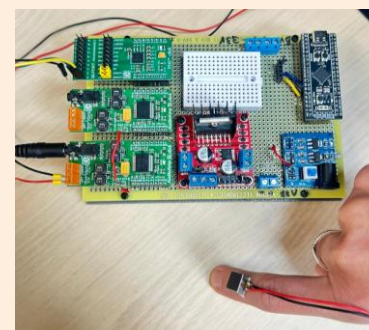
SENS3 Database



An open-source multisensory database of finger-surface interactions and corresponding sensations

Make SENS3

Experience the complete texture reproduction pipeline with a hands-on open-source hardware!



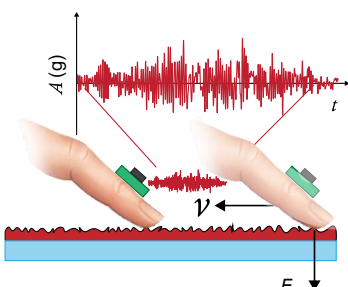
Schedule

08:30 - 09:00 Arrival and set up

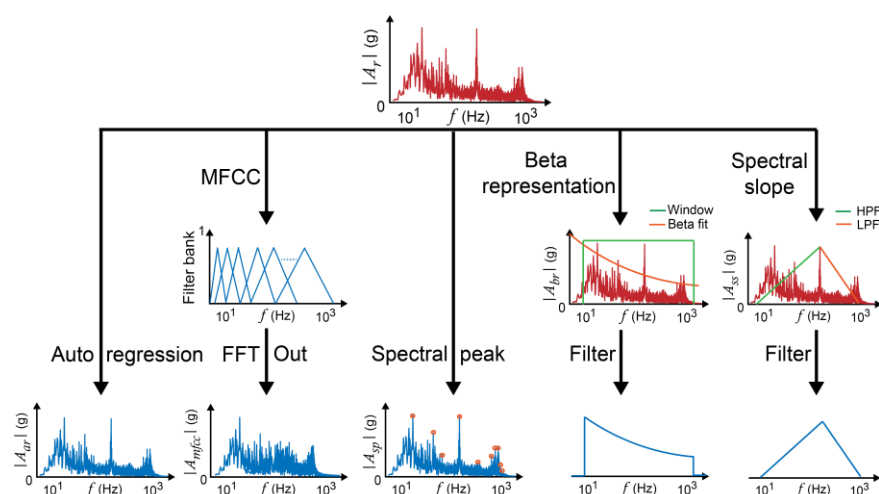
09:00 - 09:10 Introduction and overview

09:10 - 09:40 GUI download and installation

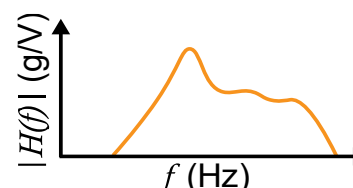
09:40 - 10:00 Texture recording and SENS3



10:00 - 10:20 Reduced texture representations



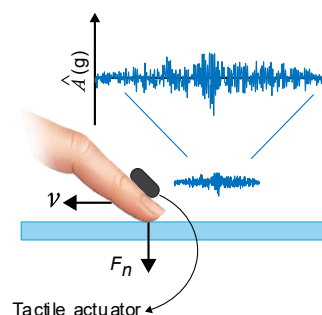
10:20 - 10:30 Actuator characterization



10:30 - 10:45 Break

10:45 - 11:30 Device compensation and inverse filtering

11:30 - 12:20 Real-time texture rendering



12:20 - 13:00 Group discussion and wrap-up

Scan to read more



Recording



Modelling



Characterisation